

AIIA Clinical Trials Dashboard: A Cloud-Native, Compliance-Embedded Clinical Trial Management System for Ayurveda Research

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Theme: MedTech / BioTech / HealthTech · Category: Software

Abstract

Ayurveda clinical research in India is still conducted largely on spreadsheets and email. Safety data is chased rather than captured, audit trails are reconstructed after the fact, and regulatory obligations under GCP-ASU, the CTRI registry and the NDCT Rules 2019 are treated as manual checklists. Off-the-shelf global Clinical Trial Management Systems (CTMS) model ICH-GCP pharma workflows but do not encode Ayurveda-specific regulation (GCP-ASU, ASU safety reporting), which makes them a poor and expensive fit for AYUSH institutions.

This paper presents the **AIIA Clinical Trials Dashboard**, a real-time, cloud-based, GCP-compliant CTMS for Ayurveda research. The system carries a study from protocol through IEC approval, CTRI registration, enrolment, visits, adverse-event (AE/SAE) capture and final reporting, exposing nine role-based views (investigator, coordinator, ethics committee, pharmacovigilance, institutional leadership, participant, regulator/auditor, and system administration roles). Regulatory rules are encoded directly into a server-side rule engine, health information is modelled with FHIR R4 / CDISC interoperability in mind, and data is stored with an immutable, time-stamped audit trail aligned to ALCOA+ principles. The prototype is deployed on a serverless edge stack (Cloudflare Workers, D1, R2) with a typed end-to-end application (TypeScript, React 19, Next.js, Drizzle).

We describe the **method** (architecture, rule encoding and evaluation), report **findings** from the running prototype (latency, payload, and operational-impact estimates), and set out **limitations** (prototype-only deployment, synthetic/benchmark data, unvalidated clinical use, integration and compliance constraints) to establish a clear, honest research agenda.

Keywords: Clinical Trial Management System, Ayurveda, AYUSH, GCP-ASU, FHIR R4, CDISC, ALCOA+, serverless, edge computing, regulatory rule engine.

1. Introduction

1.1 Background and motivation

India hosts a large and growing portfolio of Ayurveda clinical trials. According to the Clinical Trials Registry – India (CTRI), 1,392 Ayurveda trials were registered between 2018 and 2020, within a total registry of more than 107,000 trials since 2007. Despite this volume, the operational tooling for AYUSH research trails that used in mainstream pharma trials. Study teams rely on disconnected spreadsheets, shared drives and email threads for enrolment, visit scheduling, safety capture and status reporting.

This produces three recurring failure modes:

1. **Chasing, not capturing** — AE/SAE data and consent status live in inboxes, so safety signals are discovered late.
2. **Rebuilding, not reusing** — each status report is manually recompiled, because there is no single record of truth per study.
3. **Reconstructing, not auditing** — provenance is inferred after the fact, weakening ALCOA+ (Attributable, Legible, Contemporaneous, Original, Accurate) assurance.

At the same time, regulatory expectations are explicit: GCP-ASU, the CTRI registry, the NDCT Rules 2019, pharmacovigilance requirements (NPvCC), and the DPDP Act 2023 for personal data. These obligations are currently satisfied through human diligence rather than software guarantees.

1.2 Problem statement

There is no Clinical Trial Management System that understands Ayurveda regulation. Global CTMS products embed ICH-GCP assumptions and do not natively express ASU-specific safety, ethics and regulatory workflows, and their licensing cost (indicatively ₹10–15 lakh / ~\$12k–18k) is out of reach for many AYUSH institutions.

1.3 Research questions

- **RQ1 (Method):** How can Ayurveda regulatory rules (GCP-ASU, CTRI, NDCT 2019, SAE timelines) be encoded as executable, auditable software rather than documentation?
- **RQ2 (Feasibility):** Can such a system run on a zero-licence-cost, serverless edge stack with acceptable latency and footprint while remaining cloud-based and GCP-compliant?
- **RQ3 (Impact):** What operational, economic and environmental effects does a single-record, rule-driven dashboard have versus current manual practice?
- **RQ4 (Limits):** What are the boundaries of validity, and what integration, validation and adoption work remains?

1.4 Contributions

- A candidate **architecture** for an Ayurveda-aware CTMS, with compliance embedded as a server-side rule engine rather than a manual checklist.
- A **role-based data-access and KPI model** covering nine real roles in the trial lifecycle.
- A **live prototype** on a serverless edge stack with FHIR R4 / CDISC-aware data modelling and an immutable audit trail.
- A **quantified feasibility and impact assessment** with an explicit set of limitations and a phased roadmap.

2. Background and Related Work

2.1 Regulatory and standards landscape

- **GCP-ASU / ICMR-NETH (CDSR):** Good Clinical Practice for ASU trials, including ethics and safety expectations.
- **CTRI:** mandatory prospective registration of clinical trials in India.
- **NDCT Rules 2019:** statutory framework for clinical trials (India Code).

- **Ayush Suraksha / NPvCC:** ASU pharmacovigilance and adverse-event reporting.
- **DPDP Act 2023 (MeitY):** personal data protection obligations.
- **HL7 FHIR R4 and CDISC (SDTM/ADaM):** interoperability and analysis standards for health data.

2.2 Existing systems and the gap

Commercial CTMS and EDC platforms are mature for pharma ICH-GCP workflows but (a) do not encode ASU/GCP-ASU-specific semantics, (b) are priced per-seat/per-study in a range most AYUSH sites cannot justify, and (c) assume integrations (CTRI, EDC, hospital HIS APIs) that do not exist publicly. The gap is therefore not "another CTMS" but a **regulation-native, low-cost, mobile-first** system whose compliance logic is part of the software.

3. Method

3.1 Design philosophy

Three design decisions drive the method:

4. **One record per study.** A single immutable, time-stamped record is the source of truth; reports are queries over that record, not manual recompilations.
5. **Compliance as code.** Regulatory rules are executable policies evaluated on every record transition, not a checklist in a binder.
6. **Zero-training surfaces.** Role dashboards are designed so existing staff can adopt them with minimal training.

3.2 System architecture

The system follows a serverless edge architecture:

- **Client / UI:** React 19 role dashboards, mobile-first forms for visit capture.
- **Application/routing:** Next.js (App Router), with server components and API routes.
- **Language and typing:** TypeScript end-to-end, so domain models (Study, Visit, AE/SAE, Role, Consent) are statically checked.
- **Runtime:** Cloudflare Workers (global edge runtime).
- **Database:** Cloudflare D1 (SQLite) for structured trial records and the audit trail.
- **Object storage:** Cloudflare R2 for documents (consent forms, protocols, exports).
- **Schema/migrations:** Drizzle ORM for schema definition and controlled migrations.
- **Auth/crypto:** WebCrypto with PBKDF2 password hashing.
- **Interoperability:** FHIR R4 resources and CDISC (SDTM/ADaM) export mapping.

Placeholder: insert the rendered SYSTEM ARCHITECTURE diagram from Slide 3 here.

3.3 The per-record pipeline

Every record transition passes through a five-stage pipeline (from the technical approach):

7. **Capture** — visit data entered on mobile.
8. **Check** — role and consent guard applied.

9. **Apply** — safety and GCP rules evaluated.
10. **Alert** — escalation triggered against the SAE reporting clock.
11. **Report** — outputs generated for IEC, CTRI and DSUR reporting.

3.4 Encoding the rule engine

Rules are expressed as declarative policies evaluated at specific lifecycle events. Illustrative rule classes include:

Rule class	Trigger	Encoded obligation	Action
Consent guard	Visit/observation capture	Consent must be valid and unexpired	Block write; raise task
Role guard	Any read/write	Role-based access (RBAC)	Deny with audit entry
SAE clock	AE marked serious	Report within regulatory timeline	Auto-escalate; notify PV
IEC expiry	Daily/at transition	IEC approval valid for study conduct	Surface deviation/expiry
CTRI status	Enrolment open	Study must be prospectively registered	Warn/hold enrolment
ALCOA+ provenance	Any mutation	Time-stamped, attributable change	Append immutable audit entry

Each evaluation result is itself recorded, so that "why did the system allow/deny/escalate this?" is answerable as a query.

3.5 Data model and interoperability

- **Core entities:** Study, Protocol, Site, Role/User, Participant, Consent, Visit, Observation, AE/SAE, Deviation, Document, AuditEntry.
- **FHIR R4 mapping:** e.g., Patient, ResearchStudy, ResearchSubject, AdverseEvent, DocumentReference are used as the interoperability vocabulary.
- **CDISC (SDTM/ADaM):** targeted for the analysis/export phase rather than the live KPI phase.
- **Integrity:** every mutation appends an audit entry (actor, timestamp, before/after, rule outcome).

3.6 Role-based KPI model

Nine role views are defined, each with its own KPI set and least-privilege data scope, for example:

- **Investigator / Coordinator:** enrolment vs. target, visit completeness, open queries.
- **Ethics Committee (IEC):** approvals nearing expiry, deviations, amendment status.
- **Pharmacovigilance (NPvCC):** open AE/SAEs, SAE clock status, reconciliation.
- **Institutional leadership:** portfolio-level KPIs and safety signals without manual compilation.
- **Participant:** own study, visits and ID card.
- **Regulator / Auditor:** complete ALCOA+ trail as a query.
- Plus administrative/system roles.

3.7 Evaluation methodology

Because the system is a prototype, evaluation is defined as follows:

- **Performance:** cold-start latency and payload size measured on the deployed edge endpoint.
- **Functional:** end-to-end walkthroughs of a study through the five-stage pipeline, including rule triggers.
- **Impact modelling:** before/after comparison for status-reporting time, software cost, paper usage and audit-trail availability, using transparent assumptions based on the current manual baseline.
- **Compliance mapping:** each GCP-ASU / NDCT / DPDP control mapped to a software clause, with IEC sign-off as the intended governance gate.

4. Findings

*The findings below are from the **running prototype** and from transparent modelling against current manual practice. They are feasibility evidence, not results from a controlled clinical deployment; see §5.*

4.1 Technical feasibility (RQ2)

Metric	Prototype observation
Cold start	~10 ms
Payload	~31 KB
Licence cost	₹0 (open-source stack; 9 portals)
Hosting cost	₹2,500–6,000 / \$30–70 per month
Build cost	₹2.5 lakh / ~\$3,000

The edge/serverless stack met the design intent: global availability, low latency and a small footprint at near-zero licence cost.

4.2 Functional findings (RQ1)

- The five-stage pipeline (Capture → Check → Apply → Alert → Report) ran end-to-end for a study walked from protocol to report.
- Compliance was reproducibly enforced as code: role/consent guards blocked invalid writes; SAE marking triggered clock-based escalation; every mutation produced an attributable audit entry.
- Because reports are queries over the single record, no status report had to be recompiled by hand during the walkthrough.

4.3 Impact modelling (RQ3)

Before vs. after, quantified (modelled):

Indicator	Today (manual)	With dashboard	Change
Status reporting	~2 h/week by hand	Live, ~15 min/week	–87%
Software cost	₹10–15 lakh (~\$12k–18k)	₹2,500–6,000/mo	–95%

Paper per study/month	40+ printed pages	~6 pages	–85%
Audit trail	Reconstructed after the fact	Immutable, time-stamped	Always on

Monthly workload, 12-study portfolio (modelled): status reporting 100 → 24 units (–76%); deadline misses 4 → 0; manual registers 15 → 0.

Social / economic / environmental: patients gain direct visibility of their own records; ethics committees can act on deviations in days rather than months; a large lump-sum licence becomes a small monthly fee; a zero-print dashboard removes most printing.

4.4 Feasibility verdict

- **Technical:** HIGH — running live today on an open-source stack.
- **Financial:** HIGH — serviceable at ₹2,500–6,000/month with zero licence fee.
- **Market:** HIGH — demand already registered (1,392 Ayurveda trials in CTRI; 107,000+ trials; market → ₹38,900 cr by 2034, IMARC).
- **Operational:** MED–HIGH — designed for nine real roles, mobile-first, ~3 dev-days/month stewardship.

The riskiest parts already operate as a live prototype at zero licence cost. The principal open risk is **adoption**, which is budgeted for rather than assumed.

5. Limitations

We state these explicitly so the draft is defensible under review.

12. **Prototype, not a deployed product.** All results come from a running prototype and from modelling. There has been no prospective deployment inside a live trial at an AYUSH institution.
13. **Synthetic / benchmark data.** Performance figures (latency, payload) are single-environment observations, not results from a multi-site load study or production dataset.
14. **Impact estimates are modelled, not measured.** The –87% / –95% / –85% figures derive from a stated manual baseline and assumptions; they require validation in a real 12-study portfolio.
15. **No public integrations.** CTRI, hospital HIS and EDC systems expose no usable public APIs; the current approach is CSV and FHIR bulk import, with live sync deferred until an MoU. The integration risk is real and unresolved.
16. **Compliance is partly non-technical.** GCP-ASU and DPDP obligations involve legal/validation work (validation plan, SOPs, IEC sign-off). The system maps controls to clauses but has not been independently audited or validated (e.g., GAMP-5 style CSV).
17. **CDISC SDTM/ADaM is not yet complete.** Export is a large second phase; currently only FHIR-oriented interoperability and KPI/audit are delivered.
18. **No clinical-accuracy claim.** The system does not diagnose or decide; rule outcomes are process/compliance checks. Clinical correctness of entered data remains a site responsibility.
19. **Security/scale not pen-tested.** PBKDF2 auth and RBAC are implemented, but no third-party penetration test or formal threat model has been published.

20. **Adoption unproven.** Senior clinician acceptance of a new screen is the dominant open risk; zero-training design and shadowing are strategies, not yet evidence.
21. **Single-runtime dependency.** The architecture depends on one serverless vendor; despite portability intent, a vendor-outage or policy change is a continuity risk (mitigated operationally, not yet architecturally).

6. Future Work

- **Phase 1 (now):** KPI dashboards + immutable audit trail (current prototype).
- **Phase 2:** FHIR R4 integration and pharmacovigilance workflows; CSV/FHIR bulk import for CTRI/HIS/EDC.
- **Phase 3:** CDISC SDTM/ADaM export, DSUR generation, and formal validation documentation.
- **Cross-cutting:** a real-world pilot at an AIIA/AYUSH site, cognitive walkthroughs with clinicians, independent security review, and quantification of adoption KPIs.

7. Conclusion

This draft argues that the right lever for Ayurveda clinical research is not a cheaper copy of a pharma CTMS but a **regulation-native** one: compliance encoded as executable rules, one immutable record per study, nine role-based views, and a serverless edge stack with zero licence cost. The running prototype demonstrates technical and financial feasibility and a plausible operational impact, while the declared limitations define an honest, phased path from demo to validated, deployed system supporting GCP-ASU, CTRI, NDCT 2019 and DPDP obligations.

8. References and Links

- Clinical Trials Registry – India (CTRI) — ctri.nic.in
- Ayush Suraksha (pharmacovigilance) — suraksha.ayush.gov.in
- GCP / ICMR-National Ethical Guidelines (CDSR) — cdsr.nic.in
- NDCT Rules 2019 — indiacode.nic.in
- DPDP Act 2023 (MeitY) — meity.gov.in
- HL7 FHIR R4 — hl7.org/fhir
- CDISC (SDTM/ADaM) — cdisc.org
- Central Council for Research in Ayurvedic Sciences / All India Institute of Ayurveda — aiia.gov.in
- IMARC Group, India clinical trials / CTMS market projections (as cited in project materials)

Project artefacts

- Live prototype — aiia-research-observatory-demo.vercel.app
- Source code — github.com/Amirthaa1810/aiia-research-observatory
- Video demo — youtu.be/9rndwTS11ps

Status: draft for review. Figures marked "modelled" are estimates derived from a stated manual baseline and require pilot validation before being cited as measured results.